

```

libname fm93 'F:\empiricalfinance\Fama&French 1993\data';

filename odsout "F:\empiricalfinance\Fama&French 1993\result";
ods html path=odsout body="1993-Fama and French from 2003 to 2012.htm" (title="1993-Fama
and French from 2003 to 2012");

/*交易数据 allmon, 改变 WIND 回报率, 用绝对值*/
data allmon;
set fm93.monthly;
where stkcd like '60%' or stkcd like '00%' or stkcd like '300%'; /*选出所有上海和深圳 A 股*/
year=input(substr(left(Trdmnt),1,4),4.); /*提取年度*/
month=input(substr(left(Trdmnt),6,2),2.); /*提取月度*/
Msmvttl=Msmvttl*1000; /*国泰安的数据是以千为单位的*/
Msmvosd=Msmvosd*1000;
rename Mretwd=raw_return Msmvttl=marketvalue Msmvosd=amarketvalue; /*marketvalue 是总
市值, amarketvalue 是流通市值*/
run;
proc sort data=allmon; by stkcd year month; run;

data allmon1; set allmon; by stkcd;
if first.stkcd^=1 then lamarketvalue=lag(amarketvalue);else lamarketvalue=.;run;

data fm93.allmon; set allmon1; run;

/*一.构造解释变量*/
/*1、计算股票市场的三因素*/
/*首先计算 BM 指标*/
data a;
set fm93.assetdebt;
where stkcd like '60%' or stkcd like '00%' or stkcd like '300%'; /*选出所有上海和深圳 A 股*/
year=input(substr(left(Accper),1,4),4.); /*提取年度*/
month=input(substr(left(Accper),6,2),2.); /*提取月度*/
/*if year>=1996; 从 1996 年开始*/
if month=12; /*选择 12 月月末的数据（即年报值）*/
rename A003000000=bookvalue accper=date;
label accper=date;label a003000000=bookvalue;
keep stkcd accper year month A003000000; /*取每年年末的净资产数据*/
run;
proc sort data=a;
by stkcd year;
run; /*数据集 a, 包含符合条件个股的年报里的股东权益信息*/

```

```

data fm93.bookvalue; set a; run;

/*提取市值数据（包括总市值和流通市值），并与净资产数据合并*/
data a; set allmon; if month=12; run; /*市值取 12 月的*/
data a1; set a; keep stkcd year month amarketvalue marketvalue; run;
data b; set fm93.bookvalue; run;

proc sort data=a1; by stkcd year; run;
proc sort data=b; by stkcd year; run;
data c; merge a1(in=id) b; by stkcd year; if id;run;
data d; set c; bm1=bookvalue/amarketvalue; bm2=bookvalue/marketvalue; run;
data d1; set d; if bm2^='.'; if bm2>0; run; /*仅保留净资产为正的公司*/
data d2; set d1; drop date month; run;
data fm93.bm; set d2; run; /*将每年末的个股 BM 指标数据储存在 fm93.bm 中*/

data w; set fm93.bm; where stkcd like '60%'; run; /*以上海主板市场分位数为分组依据*/
proc sort data=w; by year; run;
proc univariate data=w noprint;
var bm2; by year;
output out=bm2 pctlpts=30 70 pctlpre=dec; run; /*取 30 70 分位数*/
data fm93.bm2rank; set bm2; run;
/*按照 BM 大小分为三组*/
data w1; set fm93.bm; run; /*提取 A 股所有股票进行分组*/
proc sort data=w1; by year; run;
proc sort data=fm93.bm2rank; by year; run;
data w2; merge w1(in=id) fm93.bm2rank; by year;if id; run;
data w3; set w2; if bm2<=dec30 then HL=0;
if dec30<bm2<=dec70 then HL=1; if bm2>dec70 then HL=2;
rankyear=year+1; /*rankyear 为排序的那一年，即 t 年末的 BM 作为 t+1 年 6 月份的排序标准
*/
run;
data fm93.bmrank; set w3; keep stkcd rankyear HL; run;

/*按照年度 6 月份来划分 size 的规模*/
data a; set fm93.allmon; if month=6;keep stkcd amarketvalue year; if amarketvalue^='.'; run;
data a1; set a; where stkcd like '60%'; run; /*同样以上海主板市场分位数为分组依据*/
proc sort data=a1; by year amarketvalue; run;
proc univariate data=a1 noprint;var amarketvalue; by year; output out=size pctlpts=50
pctlpre=dec; run;

proc sort data=a; by year; run; /*对 A 股所有股票进行分组*/
proc sort data=size; by year; run;

```

```

data c; merge a(in=id) size; by year; if id;run;
data c1; set c; if amarketvalue>dec50 then big=1; else big=0;
rankyear=year; /*rankyear 为排序的那一年, 即 t 年 6 月末的 Size 作为 t 年 6 月末的排序标准
*/run;

```

```

data fm93.sizerank; set c1; keep stkcd rankyear big; run;

```

```

data bm; set fm93.bmrank; run;
data sz; set fm93.sizerank; run;
proc sort data=bm; by stkcd rankyear; run;
proc sort data=sz; by stkcd rankyear; run;
data bmsize; merge bm(in=id1) sz(in=id2); by stkcd rankyear;
if id1 and id2; run;

```

```

/*生成六个组合*/
data bmsize1; set bmsize; portnum=big*3+hl; run;
proc sort data=bmsize1; by stkcd rankyear; run;
data fm93.bmsize; set bmsize1; run; /*已经按照 bm size 分组好*/

```

```

/*合并收益数据和分组数据*/
data a;
set fm93.allmon;
if month>=7 then rankyear=year;
if month<7 then rankyear=year-1;
run;
proc sort data=a; by stkcd rankyear; run;
data a1; merge a(in=id1) fm93.bmsize(in=id2); by stkcd rankyear; if id1 and id2;run;

```

```

/*计算每个组合的市值加权月度回报*/
proc sort data=a1; by Trdmnt portnum; run;
proc means data=a1 noprint; var raw_return; weight lamarketvalue; by year month portnum;
/******用流通市值加权******/
output out=portreturn mean=portreturn; run;
data portreturn1; set portreturn; where year>=2003; run;

```

```

/*计算 SMB 和 HML 因子*/
data portreturn2;
set portreturn1;
smb=(lag3(portreturn)+lag4(portreturn)+lag5(portreturn))/3-(portreturn+lag(portreturn)+lag2(po
rtreturn))/3;
hml=(portreturn+lag3(portreturn))/2-(lag2(portreturn)+lag5(portreturn))/2;
if portnum=5;
keep year month smb hml;
run;

```

```
data fm93.smb_hml; set portreturn2; if year^=2014;run;
```

```
/*全部 A 股的流通市值加权指数即 RM 的计算*/
```

```
data a; set fm93.allmon; by stkcd year month; run;
```

```
data a1; set a;
```

```
where lamarketvalue^=.; /*前一个月个股流通市值不缺省，则进入本月的流通市值加权计算*/
```

```
run;
```

```
/*删除上市六个月以内的收益数据*/
```

```
data ipo; set fm93.ipo; run;
```

```
proc sort data=ipo; by stkcd; run;
```

```
proc sort data=a1; by stkcd; run;
```

```
data a2; merge a1(in=id) ipo; by stkcd; if id; run;
```

```
data a3; set a2; ipoyear=substr(Listdt,1,4)+0; ipomonth=substr(Listdt,6,2)+0; run;
```

```
data a4; set a3; if year=ipoyear and month-ipomonth<=6 then delete;
```

```
if year-ipoyear=1 and month+12-ipomonth<=6 then delete;
```

```
run;
```

```
proc sort data=a4; by year month stkcd; run;
```

```
proc means data=a4 noprint;
```

```
var raw_return; weight lamarketvalue; /*流通市值加权平均求值*/
```

```
by year month;
```

```
output out=vwindex(drop=_freq_ _type_) mean=vwindex;
```

```
run;
```

```
data fm93.vwindex; set vwindex; where year>=2003; run;
```

```
/*2. 债券市场两因素 TREM 和 DEFAULT 计算*/
```

```
data trade; set fm93.BND_trade; run;
```

```
data a; set trade; keep _COL0 _COL1 _COL14;
```

```
rename _COL0=bndcd _COL1=trdmnt _COL14=return;
```

```
run;
```

```
data a1; set a; return=return/1200;
```

```
label return='月度化到期收益率'; run;
```

```
data inf; set fm93.bnd_bndinfo;
```

```
rename _COL0=bndcd _COL16=term _COL6=bndtype; run;
```

```
/*先计算 TERM 因素*/
```

```
data gb; set inf; where bndtype='01'; run; /*选出国债*/
```

```
proc sort data=a1; by bndcd; run;
```

```
proc sort data=gb; by bndcd; run;
```

```

data gbtrade; merge a1(in=ie) gb(in=id); by bndcd; if id and ie; run;
data b; set gbtrade; if term>10; run; /*期限大于 10 年的设为长期国债*/
proc sort data=b; by trdmnt; run;

proc means data=b noprint;
by trdmnt;
var return;
output out=lgbnd(drop=_freq_ _type_) mean=lgbnd;
run;
data lgbnd2; set lgbnd; if substr(trdmnt,1,4)+0>=2003; run;
data nrrt; set fm93.nrrt; nnrmt=nnrmt/100; run;

data term; merge lgbnd2 nrrt; by trdmnt; run;
data term1; set term; term=lgbnd-nnrmt;run;
/*TERM: 长期政府债券偏离无风险收益的程度，即每月长期政府债券回报率与无风险利率
的差*/
data fm93.term; set term1; keep trdmnt term; run;

/*计算 DEFAULT 因素*/
data firmbnd; set inf; where bndtype='02' or bndtype='03';
where _COL32='N' and _COL36='N' and _COL40='N'; run; /*剔除含权债券*/
/*选取公司债或企业债*/

proc sort data=a1; by bndcd; run;
proc sort data=firmbnd; by bndcd; run;

data fdtrade; merge a1(in=ie) firmbnd(in=id); by bndcd; if id and ie; run;
data b; set fdtrade; if term>=10; run;
proc sort data=b; by trdmnt; run;

proc means data=b noprint;
by trdmnt;
var return;
output out=fdbnd(drop=_freq_ _type_) mean=fdbnd;
run;
/*得到公司债券组合的收益率*/
data fdbnd2; set fdbnd; if substr(trdmnt,1,4)+0>=2003; run;
data fm93.cb; set fdbnd2; run;

data dft; merge fdbnd2 lgbnd2; by trdmnt; run;
data dft2; set dft; default=fdbnd-lgbnd; run;

```

/*DEF: 长期企业债券的市场组合与长期政府债券回报率的差*/

```
data fm93.default; set dft2; keep trdmnt default; run;
```

/*二.构造被解释变量*/

/*1.股票按照 Size 和 BM 构造 5*5=25 个组合*/

```
data bm; set fm93.bm; run;
```

```
data w; set bm; where stkcd like '60%'; run; /*以上海主板市场分位数为分组依据*/
```

```
proc sort data=w; by year; run;
```

```
proc univariate data=w noprint;
```

```
var bm2; by year;
```

```
output out=bm5 pctlpts=20 40 60 80 pctlpre=dec; run; /*取 30 70 分位数*/
```

/*按照 BM 大小分为五组*/

```
proc sort data=bm; by year; run;
```

```
proc sort data=bm5; by year; run;
```

```
data w1; merge bm(in=id) bm5; by year;if id; run;
```

```
data w2; set w1; if bm2<=dec20 then HL=1;
```

```
if dec20<bm2<=dec40 then HL=2;
```

```
if dec40<bm2<=dec60 then HL=3;
```

```
if dec60<bm2<=dec80 then HL=4;
```

```
if bm2>=dec80 then HL=5;
```

```
rankyear=year+1; /*rankyear 为排序的那一年，即 t 年末的 BM 作为 t+1 年 6 月份的排序标准  
*/
```

```
run;
```

```
data fm93.bm5rank; set w2; keep stkcd rankyear HL; run;
```

```
data a; set fm93.allmon; if month=6;keep stkcd amarketvalue year; if amarketvalue^=.; run;
```

```
data a1; set a; where stkcd like '60%'; run; /*同样以上海主板市场分位数为分组依据*/
```

```
proc sort data=a1; by year amarketvalue; run;
```

```
proc univariate data=a1 noprint;
```

```
var amarketvalue; by year;
```

```
output out=size pctlpts=20 40 60 80 pctlpre=dec; run;
```

```
proc sort data=a; by year; run; /*对 A 股所有股票进行分组*/
```

```
proc sort data=size; by year; run;
```

```
data c; merge a(in=id) size; by year; if id;run;
```

```
data c1; set c;
```

```
if amarketvalue<=dec20 then big=1;
```

```
if dec20<amarketvalue<=dec40 then big=2;
```

```
if dec40<amarketvalue<=dec60 then big=3;
```

```
if dec60<amarketvalue<=dec80 then big=4;
```

```
if amarketvalue>=dec80 then big=5;
```

```

rankyear=year; /*rankyear 为排序的那一年，即 t 年 6 月末的 Size 作为 t 年 6 月末的排序标准
*/
run;
data fm93.size5rank; set c1; keep stkcd rankyear big; where rankyear>=2002; run;

data bm; set fm93.bm5rank; run;
data sz; set fm93.size5rank; run;
proc sort data=bm; by stkcd rankyear; run;
proc sort data=sz; by stkcd rankyear; run;
data bmsize; merge bm(in=id1) sz(in=id2); by stkcd rankyear;
if id1 and id2; run;

/*生成 25 个组合*/
data bmsize1; set bmsize; portnum=big*5+hl; run;
proc sort data=bmsize1; by stkcd rankyear; run;
data fm93.bm5size; set bmsize1; run; /*已经按照 bm size 分组好*/

/*合并收益数据和分组数据*/
data a;
set fm93.allmon;
if month>=7 then rankyear=year;
if month<7 then rankyear=year-1;
run;
proc sort data=a; by stkcd rankyear; run;
data a1; merge a(in=id1) fm93.bm5size(in=id2); by stkcd rankyear; if id1 and id2; run;

/*计算每个组合的市值加权月度回报*/
proc sort data=a1; by Trdmnt portnum; run;
proc means data=a1 noprint; var raw_return; weight lamarketvalue; by year month portnum big
hl;
/******用流通市值加权******/
output out=portreturn(drop=_type_ _freq_) mean=portreturn; run;
data portreturn1; set portreturn; where year>=2003 and year<=2013; run;
data fm93.portreturn25; set portreturn1; run;

/*2.债券按照短期国债，长期国债，AAA、AAA-、AA+、AA、AA-及以下企业或公司债券，共
构造 7 个投资组合*/
/*因为我国债券很少有 BBB 以下的投机级别的信用评级，所以与 1993paper 略不同*/
data trade; set fm93.BND_trade; run;
data a; set trade; keep _COL0 _COL1 _COL14;
rename _COL0=bndcd _COL1=trdmnt _COL14=return;
run;
data a1; set a; return=return/1200;
label return='月度化到期收益率'; run;

```

```

data inf; set fm93.bnd_bndinfo;
rename _COL0=bndcd _COL16=term _COL6=bndtype; run;

/*短期国债组合收益率*/
data gb; set inf; where bndtype='01'; run; /*选出国债*/

proc sort data=a1; by bndcd; run;
proc sort data=gb; by bndcd; run;

data gbtrade; merge a1(in=ie) gb(in=id); by bndcd; if id and ie; run;
data b; set gbtrade; if term<10; run; /*期限小于 10 年的设为短期国债*/
proc sort data=b; by trdmnt; run;

proc means data=b noprint;
by trdmnt;
var return;
output out=lgbnd(drop=_freq__type_) mean=lgbnd;
run;
data lgbnd3; set lgbnd; if substr(trdmnt,1,4)+0>=2003; run;

/*fm93.gb_short 和 fm93.gb_long 分别记录着短期和长期的国债组合收益率*/
data fm93.gb_short; set lgbnd3; if substr(trdmnt,1,4)+0>=2003; run;
data fm93.gb_long; set lgbnd; if substr(trdmnt,1,4)+0>=2003; run;

/*计算公司或企业债组合的收益*/
data firmbnd; set inf; where bndtype='02' or bndtype='03'; rename _COL31=credit;
where _COL32='N' and _COL36='N' and _COL40='N';
run;
proc sort data=a1; by bndcd; run;
proc sort data=firmbnd; by bndcd; run;
data fdtrade; merge a1(in=ie) firmbnd(in=id); by bndcd; if id and ie; run;
proc sort data=fdtrade; by trdmnt credit; run;

proc means data=fdtrade noprint;
by trdmnt credit;
var return;
output out=firmport(drop=_freq__type_) mean=firmport;
run;
data firmport2; set firmport; if substr(trdmnt,1,4)+0>=2003; run;
/*由于中国公司（企业）债券信用评级没有低于 AA-的，所以我们暂且将其分为四组，AAA、
AA+、AA 和 AA-*/
data fm93.firmport_AAA; set firmport2; where credit='AAA'; if

```

```

substr(trdmnt,1,4)+0>=2003;run;
data fm93.firmport_AA1; set firmport2; where credit='AA+'; if
substr(trdmnt,1,4)+0>=2003;run;
data fm93.firmport_AA2; set firmport2; where credit='AA'; if
substr(trdmnt,1,4)+0>=2003;run;
data fm93.firmport_AA3; set firmport2; where credit='AA-'; if
substr(trdmnt,1,4)+0>=2003;run;

```

```

/*****Table

```

```

1*****/

```

```

data bm5size; set fm93.bm5size; rename rankyear=year; run; /*导入分组结果*/

```

```

data bm; set fm93.bm; keep stkcd year bm2; run; /*将每年末的个股 BM 指标数据储存在
fm93.bm 中*/

```

```

/*每年六月末的 marketvalue 数据*/

```

```

data size; set fm93.allmon; if month=6; keep stkcd amarketvalue year; if amarketvalue^=.;
run;

```

```

/*marketvalue 占比*/

```

```

proc sort data=size out=size2; by year; run;

```

```

data totalsize; set size2; by year; retain tmv 0;

```

```

if first.year then tmv=0; tmv=tmv+amarketvalue; if last.year;

```

```

keep year tmv; run;

```

```

proc sort data=bm5size; by year stkcd; run;

```

```

proc sort data=size; by year stkcd; run;

```

```

data size3; merge size bm5size(in=id); by year stkcd; if id; run;

```

```

proc sort data=size3; by year big hl; run;

```

```

data size4; set size3; by year big hl;

```

```

retain pmv 0; if first.year or first.big or first.hl then pmv=0;

```

```

pmv+amarketvalue; if last.year or last.big or last.hl; drop stkcd amarketvalue; run;

```

```

proc sort data=size4; by year; run;

```

```

data size5; merge size4(in=id) totalsize; by year; p=pmv/tmv*100;

```

```

if id; /*in percent*/ run;

```

```

data size; set size; amarketvalue=amarketvalue/1000000; run; /*单位: millions*/

```

```

/*计算 E/P 和 RM*/

```

```

%macro ratiofactor(infile=,var=,factor=);

```

```

data finance1; set &infile; keep stkcd year &var; run;

```

```

data finance2; set finance1; where &var^=.; run;

```

```

data proxy1; set fm93.allmon;

```

```

if month=12; keep stkcd amarketvalue year; if amarketvalue^=.; run;

```

```

data proxy2;
merge finance2(in=id) proxy1;
by stkcd year;
if id;
&factor = &var/amarketvalue*100; /*in percent*/
if &factor ^= .; /*计算指标，且指标不能缺省（缺省值来自于 12 月市值缺省）*/
keep stkcd year &factor;
where year>=2003;
run; /*数据集 proxy2，包含计算的指标最原始数据*/

data proxy3; set proxy2;
where &factor > 0;
run; /*数据集 proxy4，仅提取正的指标*/
proc sort data=proxy2 out=&factor._all; by year stkcd; run;
proc sort data=proxy3 out=&factor; by year stkcd; run;
/*若仅需要正值，则输出数据集为&factor; 若需要全值，则输出数据集为&factor._all*/
%mend;

/*计算指标：EP_positive*/
%ratiofactor(infile=fm93.fmreg,var=netprofit,factor=ep);
/*计算指标：RM，revenue 是营业收入*/
%ratiofactor(infile=fm93.fmreg,var=revenue,factor=rm);

/*分组每年求平均*/
%macro descrip(infile=,var=);
data a; set &infile;
where year>=2003 and year<=2012; /*样本统计量时间段*/
run;
proc sort data=a; by stkcd year; run;
proc sort data=bm5size; by stkcd year; run;
data fenzu; merge a(in=id1) bm5size(in=id2); by stkcd year;
if id1 and id2; run;
proc sort data=fenzu; by big hl year; run;

proc means data=fenzu noprint;
by big hl year;
var &var;
output out=ave&var(drop=_freq_ _type_) mean=ave&var;
run;
proc means data=ave&var noprint;
by big hl;
var ave&var;
output out=ave_&var(drop=_freq_ _type_) mean=ave_&var;
run;
proc transpose data=ave_&var out=transave_&var(drop=_LABEL_ _NAME_) PREFIX=BM_; by big;

```

```

id hl;run;
%mend;

/*Size 均值*/
%descrip(infile=size,var=amarketvalue);
/*BM 均值*/
%descrip(infile=bm,var=bm2);
/*E/P 均值*/
%descrip(infile=ep_all,var=ep);
/*RM 均值*/
%descrip(infile=rm_all,var=rm);

/*规模占比均值*/
proc sort data=size5; by big hl; run;
proc means data=size5 noprint;
var p;
by big hl;
output out=size6(drop=_type_ _freq_) mean=ave_p;
run;
proc transpose data=size6 out=transave_p(drop=_LABEL_ _NAME_) PREFIX=BM_; by big; id
hl;run;

/*各组年均平均公司个数*/
proc sort data=bm5size; by big hl year; run;
proc means data=bm5size noprint n; by big hl year; output out=n n=n;run;
proc means data=n noprint; by big hl; var n; output out=ave_n(drop=_freq_ _type_)
mean=ave_n;run;
proc transpose data=ave_n out=transave_n(drop=_NAME_) PREFIX=BM_; by big; id hl;run;
title "Table 1(Panel 1-Panel 5)";
title2 "Descriptive staticstics for 25 stock portfolios formed on size and book-to-market
equity:2003-2012, 10 years";
/*做出 Table 1*/
%macro print_table1(infile=,title=);
* table1 panel1 print;
title3 "&title";
proc print data=&infile noobs;
format bm_1 bm_2 bm_3 bm_4 bm_5 comma20.2;
run;
%mend;

%print_table1(infile=transave_amarketvalue,title=Panel 1: Average of annual averages of firm
size-in millions);
%print_table1(infile=transave_bm2,title=Panel 2: Average of annual B/E ratios for portfolio);

```

```
%print_table1(infile=transave_p,title=Panel 3: Average of annual percent of market value in
portfolio);
%print_table1(infile=transave_n,title=Panel 4: Average of annual number of firms in portfolio);
%print_table1(infile=transave_ep,title=Panel 5: Average of annual E/P ratios(in percent) for
portfolio);
/*%print_table1(infile=transave_rm,title=Panel 6: Average of annual RM ratios(in percent) for
portfolio);*/
```

```
/******Table
2******/
```

```
/*导入已经算好的数据*/
```

```
data nrrt; set fm93.nrrt; nnrmt=nnrmt/100; label nnrmt="月度化无风险收益率";
year=substr(trdmnt,1,4)+0; month=substr(trdmnt,6,2)+0; run;
```

```
/*短期国债组合收益率*/
```

```
data gb_short; merge fm93.gb_short nrrt; by trdmnt; gb_short=lgbnd-nnrmt;run;
```

```
/*长期国债组合收益率*/
```

```
data gb_long; merge fm93.gb_long nrrt; by trdmnt; gb_long=lgbnd-nnrmt;run;
```

```
/*计算公司或企业债组合的收益*/
```

```
data firmport_AAA; merge fm93.firmport_AAA nrrt; by trdmnt; firmport_AAA=firmport-nnrmt;
drop credit;run;
```

```
data firmport_AA1; merge fm93.firmport_AA1 nrrt; by trdmnt; firmport_AA1=firmport-nnrmt;
drop credit;run;
```

```
data firmport_AA2; merge fm93.firmport_AA2 nrrt; by trdmnt; firmport_AA2=firmport-nnrmt;
drop credit;run;
```

```
data firmport_AA3; merge fm93.firmport_AA3 nrrt; by trdmnt; firmport_AA3=firmport-nnrmt;
drop credit;run;
```

```
data term; set fm93.term; run;
```

```
data default; set fm93.default; run;
```

```
data smb_hml; merge fm93.smb_hml(in=id) nrrt; by year month; if id; keep trdmnt smb
hml;run;
```

```
data Vwindex; set fm93.Vwindex; if year<2014; run;
```

```
data r_m; merge Vwindex nrrt; by year month; rm=vwindex-nnrmt;keep trdmnt rm; run;
```

```
data cb; set fm93.cb; run;
```

/******RMRF 和四因素回归之后的 RMO 的计算******/

```
data fct; merge r_m smb_hml term default; by trdmnt; run;
proc reg data=fct noprint outest=fctreg_result rsquare adjrsq cp ;
model rm=smb hml term default/ dw spec;
output out=out2 r=r p=p l95=l u95=u;/*out2 中保留着残差*/
run;
quit;
```

```
proc sql;
create table rmo as select
out2.trdmnt,out2.r, fctreg_result.intercept
from out2, fctreg_result
;
quit;
data RMO1; set RMO; RMO=intercept+r;keep trdmnt rmo; run;
data fm93.rmo; set rmo1; run;
```

```
data sum; delete; run;
%macro Summary(var=,name=,infile=);/*输入参数分别是变量名和数据集*/
data variable; set &infile; &var=&var*100; run; /*in percent*/
proc means data=variable noprint t std;
var &var;
output out=a(drop=_type__freq_) mean=mean t=t std=std;
run;
data b; set a;
name=input("&name", $8.);
label ave=mean t=t std=std;
run;
data sum; set sum b; run;
%mend;
```

/*解释变量*/

```
%Summary(var=vwindex,name=RM,infile=Vwindex);/*股票市场指数*/
%Summary(var=gb_short,name=TB,infile=gb_short);/*短期政府债券*/
%Summary(var=gb_long,name=LTG,infile=gb_long);/*长期政府债券*/
%Summary(var=fdbnd,name=CB,infile=cb);/*长期企业债券*/
```

```
%Summary(var=rm,name=RM-RF,infile=r_m);/*股票市场超额收益*/
%Summary(var=rmo,name=RMO,infile=rmo1);/*RMRF 回归残差和截距项之和*/
%Summary(var=smb,name=SMB,infile=smb_hml);/*SMB*/
%Summary(var=hml,name=HML,infile=smb_hml);/*HML*/
%Summary(var=term,name=TERM,infile=term);/*期限因素*/
%Summary(var=default,name=DEF,infile=default);/*违约因素*/
```

```

/*债券市场被解释变量*/
%Summary(var=gb_short,name=TB,infile=gb_short);/*短期期政府债券*/
%Summary(var=gb_long,name=LTG,infile=gb_long);/*长期政府债券*/

%Summary(var=firmport_AAA,name=AAA,infile=firmport_AAA);/*AAA 级企业（公司）债券*/
%Summary(var=firmport_AA1,name=AA+,infile=firmport_AA1);/*AA+级企业（公司）债券*/
%Summary(var=firmport_AA2,name=AA,infile=firmport_AA2);/*AA 级企业（公司）债券*/
%Summary(var=firmport_AA3,name=AA-,infile=firmport_AA3);/*AA-级企业（公司）债券*/
data sum; retain name mean t std; set sum; run;

* table2 panel1 print;
    title1 "Table 2";
    title2 "Panel 1: Average t_value and standard error";
    proc print data=sum noobs;
        format _numeric_ comma20.2;
    run;

/*Autocorr. for lag*/
data corr1; delete; run;
data corr2; delete; run;
data corr12; delete; run;

%macro autocorr(var=,name=,infile=);
data variable; set &infile; &var=&var*100; run; /*in percent*/
%do t=1 %to 12;
data v; set variable; lag&t&var=lag&t(&var); run;
proc corr data=v outp=a noprint; /*相关系数*/
    var &var lag&t&var;
run;
data b; retain name lag&t&var;
    set a; if _type_=input('CORR',$8.) and &var=1;
    name=input("&name",$8.);
    keep name lag&t&var;
    rename lag&t&var=lag&t;
run;
data corr&t; set corr&t b;
run;
%end;
%mend;

%autocorr(var=rm,name=RM-RF,infile=r_m);
%autocorr(var=gb_short,name=TB,infile=gb_short);
%autocorr(var=fdbnd,name=CB,infile=cb);
%autocorr(var=rmo,name=RMO,infile=rmo1);

```

```

%autocorr(var=smb,name=SMB,infile=smb_hml);
%autocorr(var=hml,name=HML,infile=smb_hml);
%autocorr(var=term,name=TERM,infile=term);
%autocorr(var=default,name=DEF,infile=default);

%autocorr(var=gb_short,name=TB,infile=gb_short);/*短期政府债券*/
%autocorr(var=gb_long,name=LTG,infile=gb_long);/*长期政府债券*/

%autocorr(var=firmport_AAA,name=AAA,infile=firmport_AAA);/*AAA 级企业（公司）债券*/
%autocorr(var=firmport_AA1,name=AA+,infile=firmport_AA1);/*AA+级企业（公司）债券*/
%autocorr(var=firmport_AA2,name=AA ,infile=firmport_AA2);/*AA 级企业（公司）债券*/
%autocorr(var=firmport_AA3,name=AA-,infile=firmport_AA3);/*AA-级企业（公司）债券*/

data corr1; set corr1; rank=_n_; run;
data corr2; set corr2; rank=_n_; run;
data corr12; set corr12; rank=_n_; run;
data autocorr; merge corr1 corr2 corr12; by rank; drop rank; run;

*Table2 panel2 print;
title2 "Panel 2: Autocorrelation";
proc print data=autocorr noobs;
    format _numeric_ comma20.2;
run;

/*RM-RF,RMO,SMB,HML,TERM 和 DEF 之间的相关系数*/
data fct1; merge fct rmo1; by trdmnt; run;
proc corr data=fct1 outp=corrout noprint;
var rm rmo smb hml term default;
run;
data corr; set corrout; if _TYPE_=input('CORR',$8.); run;

*Table2 Panel3 print;
title2 "Panel 3: Correlation";
proc print data=corr;
    format _numeric_ comma20.2;
run;

/*股票市场 25 个投资组合的月度超额收益率的统计量*/
data a; set fm93.portreturn25; run;
data portreturn25; merge a(in=id) nrrt; by year month; portreturn=portreturn-nnrmt; if id;
rename portreturn=prex;label portreturn='组合每月超额收益率';drop nnrmt;run;

data port; set portreturn25; prex=prex*100;/*in percent*/ run;
proc sort data=port; by big hl; run;

```

```

proc means data=port noprint t std;
by big hl;
var prex;
output out=port1(drop=_type__freq_) mean=mean t=t std=std;
run;
data port2; set port1; label t="t" std="std"; run;
proc transpose data=port2 out=port3(drop=_label_) PREFIX=BM_;
by big; id hl; run;
proc sort data=port3; by _NAME_ big; run;
data port4; set port3; format bm_1 bm_2 bm_3 bm_4 bm_5 10.2; run;
data port5; set port4; big1=put(big,$1.);drop big;run;
data port6; retain big; set port5; big=input(big1,$8.);
if big="1" then big="Small"; if big="5" then big="Big"; drop big1;run;

```

```

*Table 2 Panel 4;
title2 "Panel 4: Excess returns on 25 stock portfolios formed on ME and BE/ME";
proc print data=port6 noobs;
format _numeric_ comma20.2;
run;

```

```

/*****Table
3-8*****/
/*被解释变量 Stock*/

```

```

data stk; merge portreturn25 term default r_m smb_hml rmo1; by trdmnt; run;
proc sort data=stk; by portnum; run;

```

```

/*构造一个股票市场回归和输出的宏*/
%macro stkreg(var1=,var2=,var3=,var4=,var5=,sign=,num=);
title "&sign.";
ods trace on/label listing;
proc reg data=stk noprint outest=result1 rsquare adjrsq cp tableout;
model prex = &var1 &var2 &var3 &var4 &var5/r clm cli stb ;
by portnum big hl;/*每一个分组做回归*/
output out=out1 r=r p=p l95=l u95=u ;
run;
quit;
ods trace off;

```

```

/*把截距项记录下来*/
data &sign._a; set result1;

```

```

keep portnum big hl Intercept;
x=input('PARMS',$8.);
if _TYPE_=x;
run;
/*把截距项对应的 t 值记录下来*/
data &sign._a_t; set result1;
keep portnum big hl Intercept;
x=input('T',$8.);
if _TYPE_=x;
run;

/*把残差项记录下来*/
data &sign._u; set out1; keep big hl trdmnt r; run;

/*计算 residual standard error(残差标准误)*/
data &sign._se; set result1;
se=_RMSE_*100;/*in percent*//*(_EDF+_P/_EDF_)^0.5*/;
keep portnum big hl se;
x=input('PARMS',$8.);
if _TYPE_=x;
run;

data &sign; set result1; run;

/*提取 R^2(调整后的)*/
data &sign._R2; set result1;keep portnum big hl _ADJRSQ_; x=input('PARMS',$8.);if _TYPE_=x;
rename _ADJRSQ_=R2; run;

%do n=1 %to &num;
data &sign._paramt_&&var&n; set result1; keep portnum big hl &&var&n;
x=input('PARMS',$8.);if _TYPE_=x; run;
data &sign._t_&&var&n; set result1; keep portnum big hl &&var&n; x=input('T',$8.);if
_TYPE_=x; run;

proc transpose data=&sign._paramt_&&var&n(drop=portnum)
out=&sign._paramt_&&var&n/*(drop=_LABEL_ _NAME_)*/ PREFIX=BM_;
by big; id hl; run;
proc transpose data=&sign._t_&&var&n(drop=portnum)
out=&sign._t_&&var&n PREFIX=BM_;
by big; id hl; run;

/*变换 Size 的名称*/
data &sign._paramt_&&var&n; retain size;set &sign._paramt_&&var&n;
big1=put(big,$1.);size=input(big1,$8.);

```

```

    if size="1" then size="Small"; if size="5" then size="Big"; drop big1 big; run;
data      &sign._t_&&var&n;      retain      size;set      &sign._t_&&var&n;
big1=put(big,$1.);size=input(big1,$8.);
    if size="1" then size="Small"; if size="5" then size="Big"; drop big1 big; run;

```

```

proc print data=&sign._paramt_&&var&n(drop=_name_) NOOBS;
    title2 "paramt(&&var&n)";
    format BM_1 BM_2 BM_3 BM_4 BM_5 comma20.2;
run;
proc print data=&sign._t_&&var&n(drop=_name_) NOOBS;
    title2 "t(&&var&n)";
    format BM_1 BM_2 BM_3 BM_4 BM_5 comma20.2;
run;
%end;

```

```

/*R 方*/
proc transpose data=&sign._R2(drop=portnum)
    out=&sign._R2 /*(drop=_LABEL_ _NAME_)* / PREFIX=BM_ ;
    by big; id hl; run;
data &sign._R2; retain size;set &sign._R2; big1=put(big,$1.);size=input(big1,$8.);
    if size="1" then size="Small"; if size="5" then size="Big"; drop big1 big; run;
proc print data=&sign._R2(drop=_LABEL_ _name_) NOOBS;
    title2 "R^2";
    format BM_1 BM_2 BM_3 BM_4 BM_5 comma20.2;
run;

```

```

/*残差标准误*/
proc transpose data=&sign._se(drop=portnum)
    out=&sign._se PREFIX=BM_ ;
    by big; id hl; run;
data &sign._se; retain size;set &sign._se; big1=put(big,$1.);size=input(big1,$8.);
    if size="1" then size="Small"; if size="5" then size="Big"; drop big1 big; run;

```

```

proc print data=&sign._se(drop=_name_) NOOBS ;
    title2 "s(e)";
    format BM_1 BM_2 BM_3 BM_4 BM_5 comma20.2;
run;

```

```

%mend;

```

```

%stkreg(var1=term,var2=default,sign=Table3_Stock,num=2);
%stkreg(var1=rm,sign=Table4_Stock,num=1);
%stkreg(var1=smb,var2=hml,sign=Table5_Stock,num=2);
%stkreg(var1=rm,var2=smb,var3=hml,sign=Table6_Stock,num=3);

```

```

%stkreg(var1=rm,var2=smb,var3=hml,var4=term,var5=default,sign=Table7_Stock,num=5);
%stkreg(var1=rmo,var2=smb,var3=hml,var4=term,var5=default,sign=Table8_Stock,num=5);

/*债券市场的回归*/
data bnd;
    merge gb_short gb_long firmport_AAA firmport_AA1 firmport_AA2 firmport_AA3 term
    default r_m smb_hml rmo1;
    by trdmnt; run;

/*构造一个债券市场回归和输出的宏*/
%macro bndreg(var1=,var2=,var3=,var4=,var5=,sign=,num=);
title "&sign.";
ods trace on/label listing;
proc reg data=bnd noprint outest=result1 rsquare adjrsq cp tableout;
model gb_short = &var1 &var2 &var3 &var4 &var5/r clm cli stb ;
output out=out1(keep=trdmnt r1) r=r1;
model gb_long = &var1 &var2 &var3 &var4 &var5/r clm cli stb ;
output out=out2(keep=trdmnt r2) r=r2;
model firmport_AAA = &var1 &var2 &var3 &var4 &var5/r clm cli stb ;
output out=out3(keep=trdmnt r3) r=r3;
model firmport_AA1 = &var1 &var2 &var3 &var4 &var5/r clm cli stb ;
output out=out4(keep=trdmnt r4) r=r4;
model firmport_AA2 = &var1 &var2 &var3 &var4 &var5/r clm cli stb ;
output out=out5(keep=trdmnt r5) r=r5;
model firmport_AA3 = &var1 &var2 &var3 &var4 &var5/r clm cli stb ;
output out=out6(keep=trdmnt r6) r=r6;
run;
quit;
ods trace off;

data &sign; set result1; run;

/*把截距项以及对应的 t 值记录下来*/
data &sign._a; set result1;
keep _DEPVAR_ _TYPE_ Intercept;
x=input('PARMS',$8.);
y=input('T',$8.);
if _TYPE_=x or _TYPE_=y;
run;

/*把残差项记录下来*/
data &sign._u; merge out1 out2 out3 out4 out5 out6; by trdmnt; run;

```

```

/*计算 residual standard error(残差标准误)和 R^2(调整后的)*/
data &sign._R2se; set result1;
se=_RMSE_*100;/*in percent*/ /*(_EDF+_P/_EDF)^0.5*/;
keep _DEPVAR_se _ADJRSQ_;
x=input('PARMS',$8.);
if _TYPE_=x;
rename _ADJRSQ_=R2;
run;
proc transpose data=&sign._R2se out=&sign._R2se(drop=_label_); id _DEPVAR_; run;

/*回归系数和 t 值*/
data &sign._para_t; set result1; keep _DEPVAR_ _TYPE_ &var1 &var2 &var3 &var4 &var5;
x=input('PARMS',$8.); y=input('T',$8.);if _TYPE_=x or _TYPE_=y; run;
proc sort data=&sign._para_t; by _type_; run;
proc transpose data=&sign._para_t out=&sign._para_t; by _type_; id _DEPVAR_; run;

data a;set &sign._para_t; by _type_; retain rank;if first._type_ then rank=0; rank=rank+1; run;
proc sort data=a ; by rank _type_; run;/*按照特定的变量顺序进行排序*/

data &sign; set a(drop=rank) &sign._R2se; run;
*&sign;
proc print data=&sign NOOBS;
format _numeric_ comma20.2;
run;
%mend;

%bndreg(var1=term,var2=default,sign=Table3_Bond,num=2);
%bndreg(var1=rm,sign=Table4_Bond,num=1);
%bndreg(var1=smb,var2=hml,sign=Table5_Bond,num=2);
%bndreg(var1=rm,var2=smb,var3=hml,sign=Table6_Bond,num=3);
%bndreg(var1=rm,var2=smb,var3=hml,var4=term,var5=default,sign=Table7_Bond,num=5);
%bndreg(var1=rmo,var2=smb,var3=hml,var4=term,var5=default,sign=Table8_Bond,num=5);

/*****Table
9*****/
/*print 回归的截距项*/
%macro stock_a;/*股票市场*/
title "Table 9:Stock";
%do t=3 %to 7;
/*截距项*/
data a; set table&t._stock_a;/* Intercept=Intercept*100;*/run;
proc transpose data=a(drop=portnum) out=b prefix=BM_; by big; id hl; run;

```

```

data c; retain size;set b; big1=put(big,$1.);size=input(big1,$8.);
  if size="1" then size="Small"; if size="5" then size="Big"; drop big1 big; run;
proc print data=c(drop=_name__label_) NOOBS ;
  title2 "a&t";
  format BM_1 BM_2 BM_3 BM_4 BM_5 comma20.2;
run;

```

/*截距项的 t 值*/

```

data a; set table&t._stock_a_t; run;
proc transpose data=a(drop=portnum) out=b prefix=BM_; by big; id hl; run;
data c; retain size;set b; big1=put(big,$1.);size=input(big1,$8.);
  if size="1" then size="Small"; if size="5" then size="Big"; drop big1 big; run;
proc print data=c(drop=_name__label_) NOOBS ;
  title2 "a&t._t";
  format BM_1 BM_2 BM_3 BM_4 BM_5 comma20.2;
run;
%end;
%mend;
%stock_a;

```

%macro bond_a; /*债券市场*/

```

title "Table 9:Bond";
data table9; delete; run;
%do t=3 %to 7;
/*截距项和 t 值*/
data a; set table&t._bond_a; run;
proc sort data=a; by _type_;run;
proc transpose data=a out=b(drop=_name__label_); by _type_; id _DEPVAR_; run;
data c; retain lab;set b; lab="Regression %eval(&t-2)"; run;
data table9; set table9 c;
%end;
proc print data=table9 NOOBS ;
  format _numeric_ comma20.2;
run;
%mend;
%bond_a;

```

/******Table10*****

```

*****/
title "Table 10";

data fer; delete; run;
%macro janreg1(infile=,var=);/*Five-factor explanatory returns*/
data a; set &infile; if substr(trdmnt,6,2)+0=1 then jan=1;else jan=0;run;
ods trace on/label listing;
proc reg data=a noprint outest=result1 rsquare adjrsq cp tableout;
model &var = jan /r clm cli stb ;
output out=out1 r=r;
run;
quit;
ods trace off;
data b; set result1; keep _DEPVAR_ intercept jan _ADJRSQ_;
if _TYPE_=input('PARMS',$8.);rename intercept=a jan=b; label intercept=a jan=b;run;
data c; set result1; keep _DEPVAR_ intercept jan ;
if _TYPE_=input('T',$8.); rename intercept=t_a jan=t_b; label intercept=t(a) jan=t(b);run;
data d; retain _DEPVAR_ a b t_a t_b _ADJRSQ_;merge b c; by _DEPVAR_ ; run;
data fer; set fer d; run;
%mend;
%janreg1(infile=fct,var=rm);
%janreg1(infile=rmo1,var=rmo);
%janreg1(infile=fct,var=smb);
%janreg1(infile=fct,var=hml);
%janreg1(infile=fct,var=term);
%janreg1(infile=fct,var=default);

*Table 10 Panel 1:Five-factor explanatory returns;
Title2 "Panel 1:Five-factor explanatory returns";
proc print data=fer NOOBS ;
format _numeric_ comma20.2;
run;

%macro janreg2(infile=,var=,outfile=);/*Stock returns*/
data a; set &infile; if substr(trdmnt,6,2)+0=1 then jan=1;else jan=0;run;
proc sort data=a; by big hl; run;
ods trace on/label listing;
proc reg data=a noprint outest=result1 rsquare adjrsq cp tableout;
by big hl;
model &var = jan /r clm cli stb ;
output out=out1 r=r;
run;
quit;
ods trace off;

```

```

data b; set result1; keep big hl intercept jan _ADJRSQ_;
  if _TYPE_=input('PARMS',$8.);rename intercept=a jan=b; label intercept=a jan=b;run;
data c; set result1; keep big hl intercept jan ;
  if _TYPE_=input('T',$8.); rename intercept=t_a jan=t_b; label intercept=t(a) jan=t(b);run;
data d; retain big hl a b t_a t_b _ADJRSQ_;merge b c; by big hl; run;
data &outfile; set d; run;
%mend;

%janreg2(infile=portreturn25,var=prex,outfile=jan_stock);
*Table 10 Panel 2:Excess stock returns;
Title2 "Panel 2:Excess stock returns";
proc print data=jan_stock NOOBS ;
  format a b t_a t_b _ADJRSQ_ comma20.2;
run;

/*Five-factor regression residuals*/
data u; set table8_stock_u; run;
%janreg2(infile=u,var=r,outfile=jan_stock_r);
*Table 10 Panel 3:Five-factor regression residuals;
Title2 "Panel 3:Five-factor regression residuals";
proc print data=jan_stock_r NOOBS ;
  format a b t_a t_b _ADJRSQ_ comma20.2;
run;

/*债券市场*/
data fer; delete; run;/*Excess bond returns*/
%janreg1(infile=gb_short,var=gb_short);
%janreg1(infile=gb_long,var=gb_long);
%janreg1(infile=firmpor_aa3,var=firmpor_aa3);
%janreg1(infile=firmpor_aa2,var=firmpor_aa2);
%janreg1(infile=firmpor_aa1,var=firmpor_aa1);
%janreg1(infile=firmpor_aa,var=firmpor_aa);
*Table 10 Panel 4:Excess bond returns;
Title2 "Panel 4:Excess bond returns";
proc print data=fer NOOBS ;
  format _numeric_ comma20.2;
run;

data fer; delete; run;/*Five-factor regression residuals*/
%janreg1(infile=table8_bond_u,var=r1);
%janreg1(infile=table8_bond_u,var=r2);
%janreg1(infile=table8_bond_u,var=r3);
%janreg1(infile=table8_bond_u,var=r4);
%janreg1(infile=table8_bond_u,var=r5);
%janreg1(infile=table8_bond_u,var=r6);

```

```

*Table 10 Panel 5:Five-factor regression residuals;
Title2 "Panel 5:Five-factor regression residuals";
proc print data=fer NOOBS ;
    format _numeric_ comma20.2;
run;

/*****Table11*****/
*****/
data w; set ep; run;
proc sort data=w; by year ep; run;
proc rank data=w out=eprank groups=5;
by year;
ranks ep_rank;
var ep;
run;
data w2; retain stkcd yer ep rankyear ep_rank; set eprank;
    rankyear=year+1; /*rankyear 为排序的那一年，即 t 年末的 EP 作为 t+1 年 6 月份的排序标准
*/run;
data neg_ep; set ep_all; if ep<0; rankyear=year+1; ep_rank=-1;run;
data ep_r; set w2 neg_ep; run;

data a;
set fm93.allmon;
if month>=7 then rankyear=year;
if month<7 then rankyear=year-1;
run;
proc sort data=a; by stkcd rankyear; run;
proc sort data=ep_r(drop=year); by stkcd rankyear; run;
data a1; merge a(in=id1) ep_r(in=id2); by stkcd rankyear; if id1 and id2;run;

/*计算每个组合的市值加权月度回报*/
data a2; set a1; if lamarketvalue^=.; run;
proc sort data=a2; by year month ep_rank; run;
proc means data=a2 noprint; var raw_return; weight lamarketvalue; by year month ep_rank;
/*****用流通市值加权*****/
output out=portreturn(drop=_type_ _freq_) mean=portreturn; run;
data epreturn; set portreturn; where year>=2003 and year<=2013; run;
proc sort data=epreturn; by year;run;
data epreturn1; merge epreturn nrrt; by year month; if portreturn^=.;run;
data epreturn2; set epreturn1; exr=portreturn-nnrmt;run;

data fm93.epreturn; set epreturn2; run;
proc sort data=epreturn2; by ep_rank; run;
proc means data=epreturn2 std t noprint;

```

```
by ep_rank;
var exr;
output out=epreturn3(drop=_type__freq_) mean=mean std=std t=t;
run;
```

```
Title "Table11";
```

```
title2 "Panel 1:Portfolios formed on E/P";
proc print data=epreturn3 NOOBS;
format mean std t comma20.2;
run;
```

```
proc sort data=epreturn2; by year month; run;
data epstk; merge epreturn2(in=id) r_m smb_hml; by trdmnt; if id; run;
proc sort data=epstk; by ep_rank; run;
```

```
ods trace on/label listing;
proc reg data=epstk noprint outest=result1 rsquare adjrsq cp tableout;
by ep_rank;/*每一个分组做回归*/
model exr=rm /r clm cli stb ;
output out=out1;
model exr=rm smb hml/r clm cli stb ;
output out=out2;
run;
quit;
ods trace off;
```

```
data result2; set result1; if _TYPE_=input('PARMS',$8.) or _TYPE_=input('T',$8.);
keep ep_rank _model_ _type_ intercept rm smb hml _ADJRSQ_; rename _ADJRSQ_=R2; run;
proc sort data=result2; by _model_; run;
```

```
data reg1; set result2; if _model_=input('MODEL1',$32.);drop _model_ smb hml;run;
data reg2; set result2; if _model_=input('MODEL2',$32.);drop _model_;run;
```

```
proc print data=reg1 NOOBS;
    title2 "Regression(i)";
    format intercept rm R2 comma20.2;
run;
proc print data=reg2 NOOBS;
    title2 "Regression(ii)";
    format intercept rm smb hml R2 comma20.2;
run;
```

```
ods _all_ close;  
ods html;
```